Important Questions with Answers

Chapter 3: Atmosphere and Climate
Class 9 Social Science (NCERT) - Understanding Society: India and Beyond, Part 1

A. MCQs B. Fill in the Blanks C. True/False D. Match the Following E. Assertion-Reason F. Short Answer G. Long Answer

A. Multiple Choice Questions (1 mark each)

Q1. The atmosphere is held around the Earth by:

(a) Wind
(b) Gravity
(c) Insolation
(d) Humidity

Answer: (b) Gravity

Q2. Which two gases make up most of the atmosphere?

(a) Oxygen and carbon dioxide
(b) Nitrogen and oxygen
(c) Argon and nitrogen
(d) Hydrogen and helium

Answer: (b) Nitrogen (78%) and oxygen (21%)

Q3. The amount of water vapour in the atmosphere generally ranges from:

(a) 0.1 to 0.4 per cent
(b) 5 to 10 per cent
(c) 21 per cent
(d) 78 per cent

Answer: (a) 0.1 to 0.4 per cent

Q4. The most important layer of the atmosphere, where nearly all weather occurs, is the:

(a) Stratosphere
(b) Mesosphere
(c) Troposphere
(d) Thermosphere

Answer: (c) Troposphere (average height about 12 km)

Q5. Aeroplanes prefer to fly in the stratosphere because it:

(a) Has more oxygen
(b) Is free of clouds and weather disturbances
(c) Is the hottest layer
(d) Has the most water vapour

Answer: (b) Is free of clouds and weather disturbances

Q6. Most meteors entering from space burn up in the:

(a) Troposphere
(b) Stratosphere
(c) Mesosphere
(d) Exosphere

Answer: (c) Mesosphere

Q7. Temperature decreases with altitude only in the:

(a) Troposphere and mesosphere
(b) Stratosphere and thermosphere
(c) Thermosphere and exosphere
(d) Stratosphere and mesosphere

Answer: (a) Troposphere and mesosphere

Q8. Auroras occur in the:

(a) Troposphere
(b) Stratosphere
(c) Thermosphere
(d) Mesosphere

Answer: (c) Thermosphere

Q9. Light gases like helium and hydrogen float into space from the:

(a) Exosphere
(b) Troposphere
(c) Stratosphere
(d) Mesosphere

Answer: (a) Exosphere (because of weak gravity)

Q10. The term 'climate' usually refers to average weather conditions over a period of:

(a) One day
(b) One month
(c) Thirty years or more
(d) One season

Answer: (c) Thirty years or more

Q11. An important factor affecting the distribution of temperature is:

(a) Insolation
(b) Humidity
(c) Precipitation
(d) Altitude of clouds

Answer: (a) Insolation (it decreases from the equator towards the poles)

Q12. Air always moves from:

(a) Low-pressure to high-pressure areas
(b) High-pressure to low-pressure areas
(c) Land to land only
(d) Cold to cold areas

Answer: (b) High-pressure to low-pressure areas

Q13. A wind that blows from the west is called:

(a) Westerly
(b) Easterly
(c) Southerly
(d) Northerly

Answer: (a) Westerly (winds are named after the direction they blow from)

Q14. The sea breeze blows:

(a) From land to sea at night
(b) From sea to land during the day
(c) From sea to land at night
(d) From land to sea during the day

Answer: (b) From sea to land during the day

Q15. The word 'monsoon' is derived from the Arabic word *mausim*, which means:

(a) Wind
(b) Rain
(c) Season
(d) Ocean

Answer: (c) Season

Q16. The south-west monsoon blows between:

(a) December and February
(b) June and September
(c) October and February
(d) March and May

Answer: (b) June and September

Q17. The north-east (winter) monsoon causes rainfall mainly in:

(a) Rajasthan
(b) Punjab
(c) Tamil Nadu, Andhra Pradesh and parts of Karnataka
(d) Himachal Pradesh

Answer: (c) Tamil Nadu, Andhra Pradesh and parts of Karnataka (the south-eastern coast)

Q18. Which text by Kalidasa mentions the onset of monsoon over central India?

(a) Arthashastra
(b) Meghadutam
(c) Brihatsamhita
(d) Krishiparashara

Answer: (b) Meghadutam (written around the 5th century CE)

Q19. Which Indian mission aims to make the nation 'Weather Ready' and 'Climate Smart'?

(a) Mission LiFE
(b) Mission Mausam
(c) Mission Shakti
(d) Mission Gaganyaan

Answer: (b) Mission Mausam

Q20. Which of these is a greenhouse gas?

(a) Nitrogen
(b) Argon
(c) Methane
(d) Helium

Answer: (c) Methane (others: carbon dioxide, nitrous oxide, water vapour)

B. Fill in the Blanks (1 mark each)

Q21. The height of a location above mean sea level is called _____.

Answer: altitude

Q22. The percentage of oxygen in the atmosphere is about _____ per cent.

Answer: 21

Q23. The boundary between the troposphere and the stratosphere is called the _____.

Answer: tropopause

Q24. The stratosphere contains a layer of _____ gas which filters harmful ultraviolet radiation.

Answer: ozone

Q25. The thermosphere extends from 80 km to _____ km.

Answer: 700

Q26. The northern and southern lights are called _____.

Answer: auroras

Q27. The incoming solar energy from the Sun that is intercepted by the Earth is called _____.

Answer: insolation

Q28. Rain, snow, sleet, hail and drizzle are all forms of _____.

Answer: precipitation

Q29. Atmospheric pressure is highest at _____ level.

Answer: sea

Q30. High temperature creates a _____-pressure area, which is associated with cloudy skies and wet weather.

Answer: low

Q31. The _____ breeze blows from land to sea at night.

Answer: land

Q32. The Indian Meteorological Department recognises _____ seasons in India.

Answer: four

Q33. The traditional Indian season of Varsha corresponds to the months of _____ and August (Gregorian calendar).

Answer: July

Q34. The Himalayan states experience two additional seasons: autumn and _____.

Answer: spring

Q35. The total amount of greenhouse gases released by human activities is called the _____ footprint.

Answer: carbon

C. True or False (1 mark each)

Q36. The atmosphere shields us from harmful radiation of the Sun including ultraviolet radiation.

Answer: True.

Q37. The density of air is lowest near the Earth's surface.

Answer: False. Air density is highest near the surface and decreases with altitude.

Q38. Most of the water vapour and clouds are found in the troposphere.

Answer: True.

Q39. In the thermosphere temperature rises very rapidly with altitude.

Answer: True.

Q40. We do not feel atmospheric pressure because air presses on us from all sides and our bodies exert a counter-pressure.

Answer: True.

Q41. Weather refers to the average conditions of a place over a long period.

Answer: False. That is climate; weather is the hour-to-hour and day-to-day condition of the atmosphere.

Q42. Temperature increases from the equator towards the poles.

Answer: False. Insolation, and hence temperature, decreases from the equator towards the poles.

Q43. Warm air can hold more water vapour than cold air.

Answer: True. As air gets warmer, its capacity to hold water vapour increases.

Q44. High pressure is associated with clear and sunny skies.

Answer: True.

Q45. The land breeze blows during the day when land is warmer than the sea.

Answer: False. The sea breeze blows during the day; the land breeze blows at night.

Q46. The Indian climate is broadly classified as tropical monsoon.

Answer: True.

Q47. The south-west monsoon accounts for most of India's annual rainfall.

Answer: True.

Q48. The north-east monsoon winds bring heavy rain to most of India.

Answer: False. They are cold and dry, and bring rain mainly to the south-eastern coast after crossing the Bay of Bengal.

Q49. Climate change is caused mainly by natural factors only.

Answer: False. It is caused mainly by human activities like burning fossil fuels, deforestation and industrial pollution.

Q50. The Punjab floods of 2025 were caused by both natural and human-made factors.

Answer: True.

D. Match the Following (1 mark per pair)

Q51. Match the layer of the atmosphere with its feature.

	Column A		Column B
1	Troposphere	A	Auroras and the ionosphere; reflects radio waves
2	Stratosphere	B	Very thin air; light gases escape to space
3	Mesosphere	C	Most weather phenomena occur here
4	Thermosphere	D	Meteors burn up here
5	Exosphere	E	Ozone layer is present here

Answer: 1-C; 2-E; 3-D; 4-A; 5-B

Q52. Match the element of weather with its description.

	Column A		Column B
1	Temperature	A	Movement of air from high to low pressure
2	Humidity	B	Presence of water vapour in the air
3	Precipitation	C	Weight of air on the Earth's surface
4	Atmospheric pressure	D	Degree of hotness or coolness; depends on insolation
5	Wind	E	Rain, snow, sleet, hail and drizzle

Answer: 1-D; 2-B; 3-E; 4-C; 5-A

Q53. Match the Indian season with its months.

	Column A		Column B
1	Winter	A	June to September
2	Summer or pre-monsoon	B	April to June
3	Monsoon or rainy	C	December to early April
4	Post-monsoon (retreating monsoon)	D	September-October (Gregorian)
5	Traditional Sharad season	E	October to December

Answer: 1-C; 2-B; 3-A; 4-E; 5-D

E. Assertion-Reason Questions (1 mark each)

Options: *(a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.*

Q54. Assertion (A): The atmosphere keeps the Earth warm.
Reason (R): It traps some of the Sun's energy and prevents it from escaping back into space.

Answer: (a)

Q55. Assertion (A): Aeroplanes fly in the stratosphere.
Reason (R): The stratosphere is free of clouds and other weather disturbances.

Answer: (a)

Q56. Assertion (A): Temperature decreases from the equator towards the poles.
Reason (R): Insolation decreases from the equator towards the poles.

Answer: (a)

Q57. Assertion (A): Low pressure areas are associated with clear and sunny skies.
Reason (R): In low pressure areas the air is cold and sinks.

Answer: (d) A and R are both false for low pressure. Low pressure means warm, rising air and cloudy, wet weather; high pressure means cold, sinking air and clear skies.

Q58. Assertion (A): A sea breeze blows in the afternoon.
Reason (R): Land becomes warmer than the sea, creating a low-pressure region over land.

Answer: (a)

Q59. Assertion (A): The south-west monsoon brings rainfall to India.
Reason (R): During summer the landmass heats up faster than the ocean, so moist winds blow from the ocean towards the low-pressure land.

Answer: (a)

Q60. Assertion (A): South-eastern India receives rain in winter.
Reason (R): North-east monsoon winds pick up moisture while crossing the Bay of Bengal.

Answer: (a)

Q61. Assertion (A): Climate change leads to more frequent floods and droughts.
Reason (R): Greenhouse gases like carbon dioxide and methane trap heat and raise global temperatures.

Answer: (a)

F. Short Answer Questions (2 to 3 marks each)

Q62. What is the atmosphere? Why is it important?

Answer: The atmosphere is a blanket of air surrounding the Earth, pulled down by gravity, made of a mixture of gases. It is vital for life, shields us from harmful radiation like ultraviolet rays, regulates the Earth's temperature by trapping some of the Sun's energy, and drives weather and climate.

Q63. Give the composition of the atmosphere.

Answer: Nitrogen 78%, oxygen 21%, argon 0.93%, carbon dioxide 0.04% and others 0.03% (helium, neon, krypton, xenon, ozone, hydrogen). It also has water vapour (0.1 to 0.4%) and tiny dust particles.

Q64. Name the layers of the atmosphere in order from the surface.

Answer: Troposphere, stratosphere, mesosphere, thermosphere and exosphere. They are separated by the tropopause, stratopause and mesopause.

Q65. Describe the troposphere.

Answer: It is the most important layer, with an average height of about 12 km. Temperature decreases with altitude. The air we breathe, most water vapour and clouds are here, and nearly all weather phenomena such as rain and fog occur here.

Q66. Why is the stratosphere ideal for flying aeroplanes?

Answer: It is free of clouds and other weather disturbances. It also contains the ozone layer, which filters harmful ultraviolet radiation.

Q67. What is the importance of the thermosphere?

Answer: Temperature rises rapidly as gas molecules absorb X-rays and short-wave ultraviolet radiation. It reflects radio waves back to Earth (helping radio transmission), includes the ionosphere, and auroras occur here.

Q68. What are auroras?

Answer: Colourful displays of light near the poles. Charged particles from the Sun (solar wind) are directed to the magnetic poles and make atmospheric gases glow. They are called Aurora Borealis in the Northern Hemisphere and Aurora Australis in the Southern Hemisphere.

Q69. Why do we not feel the pressure of the atmosphere?

Answer: Air presses on our bodies from all sides, and our bodies exert an equal counter-pressure from inside, so the pressures cancel.

Q70. Differentiate between weather and climate.

Answer: Weather is the hour-to-hour and day-to-day condition of the atmosphere. Climate is the sum total of weather conditions and variations over a large area for a long period, usually thirty years or more.

Q71. Name the major elements of weather and climate.

Answer: Temperature, precipitation, humidity, wind and atmospheric pressure.

Q72. What is humidity? How does temperature affect it?

Answer: Humidity is the presence of water vapour in the air. Warmer air can hold more water vapour, so humidity rises, clothes take longer to dry and sweat evaporates slowly, making us uncomfortable.

Q73. What is precipitation? Name its forms.

Answer: It occurs when air saturated with water vapour condenses and falls on Earth due to gravity. Forms: drizzle, rain, snow, sleet and hail. Rain is the most common.

Q74. How does atmospheric pressure vary with height and temperature?

Answer: Pressure is highest at sea level and falls rapidly with altitude. Warm air rises and creates low pressure (cloudy, wet weather); cold air sinks and creates high pressure (clear, sunny skies).

Q75. What is wind? How are winds named?

Answer: Wind is the movement of air from a high-pressure area to a low-pressure area. Winds are named after the direction they blow from, e.g. a west wind is westerly.

Q76. Differentiate between sea breeze and land breeze.

Answer: Sea breeze blows from sea to land during the day because land is warmer, creating low pressure over land. Land breeze blows from land to sea at night because land cools faster than the sea; its speed is low.

Q77. Name the four seasons of India as recognised by the IMD.

Answer: Winter (December to early April), summer or pre-monsoon (April to June), monsoon or rainy season (June to September) and post-monsoon or retreating monsoon (October to December).

Q78. Name the six traditional Indian seasons.

Answer: Vasanta (spring), Grishma (summer), Varsha (monsoon), Sharad (early autumn), Hemanta (late autumn) and Shishira (winter), each about two months long.

Q79. What is monsoon? How did it get its name?

Answer: Monsoon is the seasonal reversal in wind direction during a year. Arab traders named it from the Arabic word *mausim*, meaning season. Ancient sailors were the first to notice it.

Q80. How did ancient Indians predict the monsoon?

Answer: Krishiparashara and the Brihatsamhita describe methods. Krishiparashara used the positions of the Moon and Sun; Varahamihira used lunar mansions (nakshatras). Kautilya's Arthashastra recorded measurements of rainfall for revenue and relief management.

Q81. What is the National Monsoon Mission? What is Mission Mausam?

Answer: Under the National Monsoon Mission, the Ministry of Earth Sciences developed advanced weather and climate prediction models to improve monsoon prediction. Mission Mausam aims to make India a global leader in weather and climate science, 'Weather Ready' and 'Climate Smart'.

Q82. What is climate change? What are its effects?

Answer: Long-term changes in weather patterns such as temperature, rainfall and wind, caused mainly by human activities. Effects: more floods and droughts, melting glaciers, rising sea levels, loss of biodiversity and impacts on health, agriculture and livelihoods.

Q83. What is a carbon footprint? How can we reduce it?

Answer: The total amount of greenhouse gases released as a result of human activities such as energy use, transport or production of goods and services. We can reduce it by saving energy, using renewable sources, planting trees, avoiding single-use plastics and saving water.

Q84. Why does Shillong receive more rainfall than Kolkata? (Table 3.3)

Answer: Shillong has the highest annual rainfall (225.3 cm) because it lies on hills in the path of moisture-laden monsoon winds, while Kolkata gets 162.5 cm. (Mountains are a factor affecting precipitation.)

G. Long Answer Questions (5 marks each)

Q85. What is the atmosphere? Explain its composition with the help of a pie diagram.

Answer: The atmosphere is the blanket of gases surrounding the Earth, held by gravity. It supports life, protects us from harmful radiation, regulates temperature and forms weather and climate.

Composition (pie diagram):

- Nitrogen: 78%
- Oxygen: 21%
- Argon: 0.93%
- Carbon dioxide: 0.04%
- Others: 0.03% (helium, neon, krypton, xenon, ozone, hydrogen)

Besides these, there is water vapour (0.1 to 0.4%), important for clouds and precipitation, and dust particles. Composition also varies with altitude.

Q86. Describe the structure of the atmosphere with its layers.

Answer: The atmosphere is layered on the basis of changes in temperature and density with altitude. Density is highest near the surface.

1. **Troposphere:** about 12 km; temperature falls with height; contains air, water vapour, clouds; all weather occurs here; ends at the tropopause.
2. **Stratosphere:** up to 50 km; free of clouds, ideal for aeroplanes; contains the ozone layer; ends at the stratopause.
3. **Mesosphere:** up to 80 km; temperature falls with height; most meteors burn up.
4. **Thermosphere:** 80 to 700 km; temperature rises rapidly; contains the ionosphere; reflects radio waves; auroras occur.
5. **Exosphere:** uppermost layer with very thin air; helium and hydrogen escape into space due to weak gravity.

Q87. Describe the elements of weather and climate.

Answer: 1. **Temperature:** varies between day and night and across seasons; depends on insolation, which decreases from the equator to the poles.
2. **Humidity:** water vapour in the air; warm air holds more, causing discomfort.
3. **Precipitation:** condensed water vapour falling as drizzle, rain, snow, sleet or hail; affected by prevailing winds, mountains and seasons.
4. **Atmospheric pressure:** weight of air; highest at sea level; warm air creates low pressure (cloudy, wet), cold air creates high pressure (clear, sunny).
5. **Wind:** movement of air from high to low pressure; may be gentle or strong; named by the direction from which it blows. Local winds include land and sea breezes.

Q88. Explain the mechanism of the south-west monsoon and the north-east monsoon. How are they different?

Answer: South-west monsoon (summer monsoon): June to September. In summer the Indian landmass heats up faster than the surrounding oceans, creating low pressure over the subcontinent while the Indian Ocean stays cooler with high pressure. Moist winds blow from the Indian Ocean, Arabian Sea and Bay of Bengal towards the land, bringing rain. It gives most of India's annual rainfall.
North-east monsoon (winter monsoon): October to February. The landmass cools faster than the oceans, creating high pressure over land and low pressure over the seas. Cold, dry winds blow from land to sea and generally bring no rain, but after crossing the Bay of Bengal they bring rain to Tamil Nadu, Andhra Pradesh and parts of Karnataka.

Q89. Explain the importance of monsoon in the life of the people of India.

Answer: - Most of India's agriculture depends on monsoon rain for sowing and growing crops.
- A good monsoon ensures food production and water in rivers, reservoirs and wells.
- It affects daily life, transport, festivals and employment, especially in rural areas.
- Excessive rain causes floods, and weak monsoons lead to droughts.
- Therefore, the monsoon greatly influences the economy, lifestyle and livelihoods. Ancient texts like Krishiparashara, the Brihatsamhita and Kalidasa's Meghadutam show its long-standing importance.

Q90. Describe the four seasons of India.

Answer: 1. **Winter:** December to early April. Coldest months December and January; average temperature about 10-15 degrees C in the north-west, rising to 20-25 degrees C in the south-east.
2. **Summer (pre-monsoon):** April to June (up to July in the north-west). Hottest month is April in the west and south, May in the north; inland temperatures 32-40 degrees C.
3. **Monsoon (rainy):** June to September, dominated by the humid south-west monsoon, which sweeps across the country in late May or early June and begins to recede from north India at the beginning of October.
4. **Post-monsoon (retreating monsoon):** October to December; October and November are usually cloudless in the north-west.
Himalayan states also experience autumn and spring.

Q91. What is climate change? Explain its causes, effects and ways to reduce it.

Answer: Meaning: long-term change in weather patterns such as temperature, rainfall and wind.
Causes: burning fossil fuels, deforestation and industrial pollution release greenhouse gases (carbon dioxide, methane, nitrous oxide, water vapour), which trap heat and raise global temperatures.
Effects: more frequent floods and droughts, melting glaciers, rising sea levels, loss of biodiversity, and impact on health, agriculture and livelihoods, including women and children.
Solutions: reduce our carbon footprint, save energy, use renewable energy, protect forests, plant trees, reduce waste and plastics, save water and adopt sustainable lifestyles.

Q92. Write a case study of the Punjab floods of 2025: causes and effects.

Answer: In 2025 Punjab faced severe floods due to heavy monsoon rains and swelling of the Satluj, Beas and Ravi rivers.
Natural causes: very heavy monsoon rain intensified by western disturbances; rain in Himachal Pradesh and Jammu and Kashmir; rivers (Satluj, Beas, Ravi, Ghaggar) already flowing high.
Human-made causes: weak, old embankments (dhussi bandh); houses and farms built too close to rivers; silt in rivers and dams reducing capacity; late or unclear flood warnings.
Effects: loss of lives; thousands moved to relief camps; farmland and paddy damaged; poultry and dairy farms and animals lost; roads, bridges, border fences and public buildings damaged; waterborne diseases and sanitation problems.
Better flood management and preparedness are needed.

Q93. Explain the role of the atmosphere with the help of land and sea breezes and the effect of local winds on coastal regions.

Answer: Local winds like the land and sea breeze affect the weather and climate of a place and create moderate conditions in coastal regions.
Sea breeze (day): in the afternoon land becomes warmer than the sea, so a low-pressure region forms over land; the wind blows from the sea to land.
Land breeze (night): land cools faster than the sea, so the sea has relatively lower pressure; the wind blows from land to sea, but its speed is low because the temperature and pressure difference is small.
Both arise from differential heating and cooling of land and water, and air moves from high to low pressure.